

ANOVA Summary

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Experiments

Two ways to gather information:

1. Observation: Watch and record without interference.
 2. Experiment: Actively manipulate variables to test their effect on a response.
- ▶ Experiments involve control and random assignment → allow us to infer cause and effect.
 - ▶ Observational studies can only show association and not causation (confounding variables!)
 - ▶ We considered comparative experiments: compare the effects of a number of treatments

Terminology

- ▶ Treatment factor: the variable we manipulate
- ▶ Treatment levels: the levels/settings/categories to which we manipulate the treatment factor
- ▶ Treatments: the levels (if single factor) or combinations of levels (if multi-factor) of the treatment factor applied to experimental units
- ▶ Experimental units: the objects/entities/things/people to which treatments are applied
- ▶ Response variable: the variable we measure to assess the effect of the treatment factor
- ▶ Replicate: an independent observation under the same treatment

Three principles of experimental design

1. Replication: Each treatment is applied to more than one experimental unit.

Why? To estimate differences between experimental units with the same treatment (experimental error variance)

2. Randomisation: Assign treatments to experimental units at random.

Why? To eliminate bias and confounding variables.

3. Reduction of experimental error variance

Why? To increase the power of the experiment to detect differences between treatments. Less variability → easier to detect real effects.

Designs

1. CRD: Completely Randomised Design

- ▶ Treatment structure: Single factor or factorial.
- ▶ Blocking structure: None!
- ▶ Randomisation: Random assignment of treatments to experimental units.
- ▶ Balanced: equal number of observations per treatment.

2. RBD: Randomised Block Design

- ▶ Treatment structure: Single factor or factorial.
- ▶ Blocking structure: Blocking variable (covariate) is used to group experimental units into blocks.
- ▶ Randomisation: Random assignment of treatments to experimental units within each block.
- ▶ Complete and balanced: Treatments appear exactly one within a block.

SF CRD

- ▶ Factor A with a levels
- ▶ Levels are treatments and they are applied randomly to r experimental units
- ▶ $N = r \times a$ total experimental units / observations
- ▶ We want to determine whether there are differences in treatment means

One-way ANOVA model

$$Y_{ij} = \mu + A_i + e_{ij}$$

where

$i = 1, \dots, a$ (a = number of treatments)

$j = 1, \dots, r$ (r = number of replicates)

Y_{ij} = response value of the j^{th} unit receiving treatment i

μ = overall or general mean

A_i = effect of the i^{th} level of treatment factor A

e_{ij} = random error with $e_{ij} \sim N(0, \sigma^2)$

$$\sum A_i = 0$$

Assumptions

1. No outliers
2. Homogeneity of population variances
3. Normality of errors
4. Independent errors

Decomposition of sums of squares

$$Y_{ij} - \mu = (\mu_i - \mu) + (Y_{ij} - \mu)$$

$$Y_{ij} - \bar{Y} = (\bar{Y}_i - \bar{Y}) + (Y_{ij} - \bar{Y}_i)$$

$$\sum_i \sum_j (Y_{ij} - \bar{Y})^2 = \sum_i \sum_j (\bar{Y}_i - \bar{Y})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_i)^2$$

$$\sum_i \sum_j (Y_{ij} - \bar{Y})^2 = \sum_i r \times (\bar{Y}_i - \bar{Y})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_i)^2$$

$$SS_{\text{total}} = SS_{\text{between groups}} + SS_{\text{within groups}}$$

Hypotheses

With this model we can test the following hypotheses:

Null: All treatment means are equal, the treatment factor had no effect on the response.

Alternative: At least one treatment mean is different, the treatment factor did have an effect on the response.

ANOVA Table

Source	Sums of Squares (SS)	df	Means Squares (MS)	F
Treatment	$\sum_i n(\bar{Y}_i - \bar{Y})^2$	$a - 1$	$MSB = \frac{SS_A}{(a-1)}$	$\frac{MSB}{MSE}$
Residuals (Error)	$\sum_i \sum_j (Y_{ij} - \bar{Y}_i)^2$	$N - a$	$MSE = \frac{SSE}{(N-a)}$	
Total	$\sum_i \sum_j (Y_{ij} - \bar{Y})^2$	$N - 1$		

MSE

- ▶ MSE is the estimate of variation within treatment groups
- ▶ It is an average of the observed variation in treatments
- ▶ It is the estimate of the experimental error variance (σ^2)
- ▶ MSE determines the precision of our estimates:

$$Var(\hat{\mu}_i) = Var(\hat{A}_i) = \frac{\sigma^2}{n}$$

Why ANOVA?

- ▶ Doing multiple t-test would lead to an increased risk of making a Type 1 error.
- ▶ ANOVA is a global test for differences between multiple means.
- ▶ If we find evidence for differences, then we look at specific contrasts.

SF RCBD

- ▶ Treatment factor A with a levels
- ▶ Blocking factor with b levels
- ▶ Blocks are not of interest — we adjust for them to reduce variability.
- ▶ We don't have reason to believe that these two interact.
- ▶ If interaction is suspected → do not use this model.
- ▶ The a treatments are randomly assigned with each of the blocks

Two-way ANOVA without interaction

$$Y_{ij} = \mu + A_i + B_j + e_{ij}$$

- ▶ B_j (block effect) represents systematic differences due to blocks (e.g., soil type, teacher, fitness level).
- ▶ The block effect is interpreted in the same way as the treatment effect, it is the difference between block mean and the overall mean.
- ▶ Estimation as before using least square estimators (i.e. sample means)

Assumptions

1. No outliers
2. Homogeneity of population variances
3. Normality of errors
4. Independent errors
5. **Block effects and treatment effects are additive**

Decomposition of sums of squares

$$Y_{ij} = \mu + A_i + B_j + e_{ij}$$

$$Y_{ij} - \mu = (\mu_i - \mu) + (\mu_j - \mu) + (Y_{ij} - (\mu - A_i - B_j))$$

$$Y_{ij} - \bar{Y} = (\bar{Y}_i - \bar{Y}) + (\bar{Y}_j - \bar{Y}) + (Y_{ij} - (\bar{Y} - \hat{A}_i - \hat{B}_j))$$

$$\sum_i \sum_j (Y_{ij} - \bar{Y})^2 = \sum_i \sum_j (\bar{Y}_i - \bar{Y})^2 + \sum_i \sum_j (\bar{Y}_j - \bar{Y})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_i)^2$$

$$SS_{\text{total}} = SS_{\text{treatments}} + SS_{\text{blocks}} + SS_{\text{within groups}}$$

Hypotheses

Treatment effects:

Null: No difference between treatment means. All the treatment means are the same, which means the effects are all zero.

Alternative: There is a difference between treatment means. At least one the effects is not zero.

Block effects:

Null: No differences between block means. All block means are the same, which means the effects are all zero. Alternative: There is a difference between block means. At least one of the effects is not zero.

ANOVA Table

Source	SS	df	MS	F
Treatments A	$SS_A = b \sum_i (\bar{Y}_i - \bar{Y}_{..})^2$	$(a - 1)$	$\frac{SS_A}{(a-1)}$	$\frac{MS_A}{MSE}$
Blocks B	$SS_B = a \sum_j (\bar{Y}_j - \bar{Y}_{..})^2$	$(b - 1)$	$\frac{SS_B}{(b-1)}$	$\frac{MS_B}{MSE}$
Error	$SSE = \sum_{ij} (Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2$	$(a - 1)(b - 1)$	$\frac{SSE}{(a-1)(b-1)}$	
Total	$SS_{total} = \sum (Y_{ij} - \bar{Y}_{..})^2$	$ab - 1$		

Was blocking effective?

- ▶ Do a relative efficiency check by examining the F value.
- ▶ Blocking should reduce SSE.
- ▶ $F > 1$ then blocking was effective.
- ▶ $F \approx 1$, then we would have been equally well off with a CRD.

Factorial CRD

- ▶ Factor A with a levels
- ▶ Factor C with c levels
- ▶ We have $a \times c$ treatments.
- ▶ Here we are interested in the interaction between the two factors.

Two-way ANOVA model with interaction

$$Y_{ijk} = \mu + A_i + C_j + (AC)_{ij} + e_{ijk}$$

- ▶ Y_{ijk} is the k^{th} observation of the $(ij)^{th}$ treatment.
- ▶ $(AC)_{ij}$ is the interaction between the i^{th} level of A and the j^{th} level of C (single symbol and not the product of main effects)

Assumptions

1. No outliers
2. Homogeneity of population variances
3. Normality of errors
4. Independent errors

Decomposition of sums of squares

$$Y_{ijk} = \mu + A_i + C_j + (AC)_{ij} + \varepsilon_{ijk}$$

$$Y_{ijk} - \mu = A_i + C_j + (AC)_{ij} + \varepsilon_{ijk}$$

$$Y_{ijk} - \bar{Y}_{..} = (\bar{Y}_{i.} - \bar{Y}_{..}) + (\bar{Y}_{.j} - \bar{Y}_{..}) + (\bar{Y}_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..}) + (Y_{ijk} - \bar{Y}_{ij})$$

$$\begin{aligned} \sum_{i,j,k} (Y_{ijk} - \bar{Y}_{..})^2 &= \sum_i an(\bar{Y}_{i.} - \bar{Y}_{..})^2 + \sum_j bn(\bar{Y}_{.j} - \bar{Y}_{..})^2 \\ &\quad + \sum_{i,j} n(\bar{Y}_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} + \bar{Y}_{..})^2 + \sum_{i,j,k} (Y_{ijk} - \bar{Y}_{ij})^2 \end{aligned}$$

Hypotheses

Main effects A:

Null: All main effects of A are zero. The mean response did not differ between the levels of A.

Alternative: At least one the effects is not zero. The mean response did differ between the levels of A.

Main effects C:

Null: All main effects of C are zero. The mean response did not differ between the levels of C.

Alternative: At least one the effects is not zero. The mean response did differ between the levels of C.

Interaction effects:

Null: All interaction effects are zero. The factors do not interact, they're effects are additive.

ANOVA Table

Source	SS	df	MS	F
A Main Effects	$SS_A = nc \sum_i (\bar{Y}_{i..} - \bar{Y}_{...})^2$	$(a - 1)$	MS_A	$\frac{MS_A}{MS_E}$
C Main Effects	$SS_C = na \sum_j (\bar{Y}_{.j.} - \bar{Y}_{...})^2$	$(c - 1)$	MS_C	$\frac{MS_C}{MS_E}$
AC Interactions	$SS_{AC} = n \sum_{ij} (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2$	$(a-1)(c-1)$	MS_{AC}	$\frac{MS_{AC}}{MS_E}$
Error	$SS_E = \sum_{ijk} (Y_{ijk} - \bar{Y}_{ij.})^2$	$ac(n - 1)$	MSE	-
Total	$SS_{total} = \sum_{ijk} (Y_{ijk} - \bar{Y}_{...})^2$	$abn - 1$	-	-

Summary of the process

► For each design considered:

1. A model which describes the variation in our response in terms of the factors of our experiment.
2. This model gives us the decomposition of the total sums of squares (variation in our response)
3. Each SS captures a different source of variation: A, C, interaction, residual.

Summary of the process

4. For each sum of square we can divide by the degrees of freedom to get mean squares
 5. For each factor in our model, we compare its mean square to the error mean square
 6. This gives an F value, which measures how large the effects are relative to the error variance.
 7. Compare this F value to the null distribution and make a conclusion
- ANOVA uses **variation decomposition** to detect **mean differences**.

ANOVA

- ▶ Each ANOVA table can only tell about the presence of:
 1. Treatment effects in a SF CRD or SF RCBD
 2. Main effects and interaction effects in a factorial CRD
- ▶ It does not tell us which effects exist and how they differ.
- ▶ For that we need contrasts!

ANOVA tells us **if** there's a difference; contrasts tell us **where**.

- ▶ Comparing treatment means after finding evidence for a difference in means.
- ▶ Contrasts allow us to ask targeted questions (e.g. comparing specific treatments).
- ▶ Use regression parameterisation.
- ▶ Only for one-way ANOVA and two-way ANOVA without interaction.

Confused, unsure?

- ▶ That's fine! You are learning.
- ▶ Put in the work to understand.
- ▶ Consult your notes.
- ▶ Do the tutorials.
- ▶ Ask questions in the discussion board.
- ▶ Come speak to me during consultations time.
- ▶ Take care of yourself: Eat, drink water, sleep and exercise.